Long-Khac Nguyen

LongNK0702 | Long Nguyen Khac | nk.long723@gmail.com | (+84)358036759

EDUCATION

Vietnam National University, Hanoi (VNU) Hanoi, Vietnam
B.Eng. in Automation and Informatics Sep 2021 - Dec 2026

- GPA: 3.71/4.0 (Rank #3 of the total major, Top 4% of the cohort)
- Awards: Most Popular Poster-GCF(2026), TOSHIBA Scholarship (2024), Outstanding Young Face Award (2024), Student with Five Good Merits (2024), Spirit Prize-IEEE SEACAS Hackathon (2022)

RESEARCH EXPERIENCE

Research Fellow Singapore
Water & Energy Research Lab, EEE, Nanyang Technology University (NTU) Feb 2026 - Current

- **Project:** Advanced 3D Wireless Power Transfer System for Autonomous Underwater Vehicles.
  - Developed a 3D wireless power transfer system for AUVs; optimized mechanical design and communication.
  - Analysed the efficiency and output power of UWPT system in AUV prototype.

Research Intern Chiayi, Taiwan
AISC Lab, AIM-HI Institute, National Chung Cheng University May 2025 - Jul 2025

- **Project:** Smart Sensing Technology for Smart Manufacturing.
  - Built data-driven models to estimate cutting forces using 12 datasets and sparse identification.
  - **Results:** Achieved $R^2 = 0.5$–0.6.

Student Research Hanoi, Vietnam
Intelligent Control and Artificial Intelligence Lab, VNUIS Sept 2022 - Current

- **Project:** Implementation Adaptive Control for 3-DOF Parallel Robot (2024)
  - Developed a 3-DOF delta robot with adaptive fuzzy backstepping control; reduced trajectory error.
  - **Results:** Joint trajectory deviations < 0.15 rad (vs. 0.3 rad), with stability reached in 0.15–0.2 s (vs. 0.3 s).
- **Project:** Detection of Personal Protective Equipment (PPE) in Construction Sites (2023)
  - Built PPE detection system using YOLOv5s on 2,632 images.
  - **Results:** SiLU achieved Precision 84.8%, Recall 80.2%, mAP@0.5 83.9%, mAP@0.95 43.6%.

SELECTED PUBLICATIONS

- Nguyen Khac Long et al., *Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot*, ICTA 2024.
- Nguyen Khac Long et al., *YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites*, STAIS 2023.

LEADERSHIP & GLOBAL EXPERIENCE

- **Global Connect Fellowship (GCF-NTU) 2026** – Selected as 1 of 39 young scholars globally (from 2,700 applicants) for a fully funded research fellowship at Nanyang Technological University, Singapore.
- **DiscoverNUS 2025, VNU Representative** – Selected as 1 of 8 outstanding VNU students for a fully funded semester exchange program at National University of Singapore.
- **Taiwan Experience Education Program (TEEP) 2025** – Selected as the only VNU student for a fully funded research internship at National Chung Cheng University, Taiwan.
- **ASEAN in Today's World 2025, VNU Representative** – Selected as 1 of 4 VNU students for a funded program in Indonesia.
- **Temasek Foundation SCALE 2024, VNU Representative** – Selected as 1 of 20 VNU students for a fully funded leadership and sustainability exchange program in Singapore.

Skills

Technical: Python, C/C++, MATLAB, LaTeX, SolidWorks, Arduino, ESP32, Raspberry Pi, Jetson Nano
Domains: Robotics, Control Systems, AI Applications, Mechanical Design, Prototyping
Languages: Vietnamese (Native), English (IELTS 6.5)